Project TriSweep

1. Project Overview

Project TriSweep is a 3 lane vertical scrolling rhythm game with a hint of danmaku using a mouse. Players have to press to the beat while evading obstacles/bullets or collect the note to clear the chart. This game aims to give a casual player a more friendly approach to clear a song, while providing the experienced player a challenge to all perfect the chart.

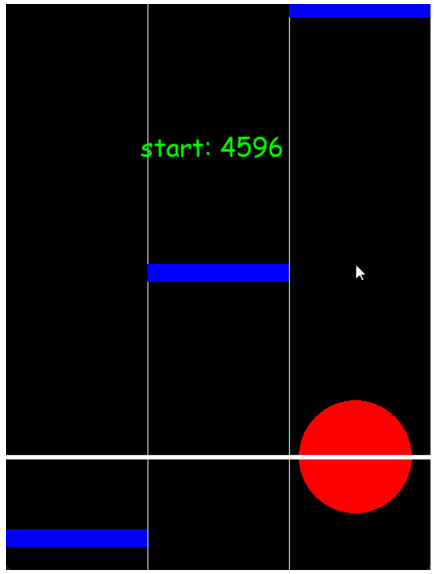
2. Project Review

This game is inspired from O.N.G.E.K.I. (オンゲキ), an arcade rhythm game that is also a danmaku. However, this game hasn't made any attempt to be released globally. Moreover, the learning curve of this game is very high due to the many mechanics of the game and the tight judgement timing.

I'm borrowing the break note and scoring mechanic from another arcade rhythm game maimai DX (maimai でらっくす). This game has 5 tiers of judgement level. Perfect judgement gives the same score as critical perfect that have tighter judgement timing except for break note, where critical perfect gives a bonus score that will be calculated in the final percentage. And the theoretical score percentage that one can get is 101%. This will help the new/casual player clear a chart while giving experienced players a goal to complete.

3. Programming Development

3.1 Game Concept

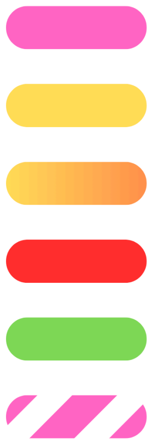


(Game prototype to see whether it is viable to play)

Project TriSweep is a 3 lane vertical scrolling rhythm game with a twist, players also have the character in the middle that have to avoid or catch the note in the play area. This game uses 6 keys and a mouse to play. The layout is as follows.

(Example of the PC game setup)

From the picture, the mouse is intended to be either above or below the keyboard. Or alternatively, you can use a laptop touchpad to control the character. The input for keyboard 3 keys split left and right. Using either side of the keyboard is fine as it is linked to the same lane on each side.



Tap - 1 note appearing in one row

Both - 2+ note appearing in one row

Break - Tap/Both with stricter window

Avoid - Avoid character from the note

Catch - Catch the note with the character

Stripe (Modifier) - Only Crit.P or Miss

(Examples of note type in the game. Subject to change)

This game will differentiate the note type with color as the picture above and also its shape.

There are 5 levels of judgement in this game as follow:

- Critical Perfect: Awards 100% of the note score
- Perfect: Awards 100% of the note score
- Great: Awards 80% of the note score
- Good: Awards 50% of the note score
- Miss: Gives no note score

Break note behaves differently in that it will give more score than normal note, and that there is bonus score for hitting critical perfect judgement.

After finishing the chart, players will be given a score for their performance. The game will calculate the normal scoring and bonus scoring separately, then combined back together. The theoretical score that can be achieved in this game is 101.0000%. 95 for the normal note score, 5 for catch note score, and 1 for bonus score.

Players can fail in the middle of the song if they collide with the avoid note too much. Additionally, a player needs to score at least 80% of the chart to be considered clear.

3.2 Object-Oriented Programming Implementation

Name	Key Attribute	Key Method	Comment
NoteBase	LaneNumber, TimeStart, TimeEnd, BaseScore, BonusScore	DrawNote, GetScore	Contains the information of the note timing, lane, and drawing.
GameManager	PlayingSong, PlayingChart, IsPaused, NoteSpeed, Dict[Note, NoteJudgement]	ProcessInput, CalculateJudgementLevel,	Contains all the functions of the game. (Non-rendering part).
Chart	DifficultyConstant, DifficultyName, ChartDesigner, List[Note]	LoadAllNotes,	Contains the information of all note placement in the chart file.
Song	SongName, Artist, BPM, List[Chart], PreviewStart, PreviewDuration	GetChartList	Contains all data from the song. Artist, BPM, List of charts, etc…
Player	Health, Position, Size	Move, Damage, Heal, DrawPlayer	An object at the judgement line, it acts as a player character that

Name	Key Attribute	Key Method	Comment
			the player needs to move them with their mouse to avoid or collect the note.
Result	CurrentSong, CurrentChart, MaxBaseRaw, MaxBonusRaw, TotalBaseRaw, TotalBonusRaw, ScorePercent, JudgementResult, HitError, KeyPressSide, Combo, PlayerPosition	RecordHitData, WritePlayRecordJson, WritePlayerRecordJson, GetSongChartName, RecordPlayerPosition	Contains the result of the play.

3.3 Algorithms Involved

Note Spawning: The game will spawn the note from the notebase data. It will try to spawn in a way that the bottom of the note will hit the judgement line.

Input Processing: GameManager will record the millisecond of the frame that player raised the keypress event. It will then compare how off the input is and give a judgement level.

4. Statistical Data (Prop Stats)

4.1 Data Features

Data Name	Why is it good to have this data? What can it be used for	How will you obtain 100 values of this feature data?	Which variable (and which class will you collect this from?)	How will you display this feature data (via summarization statistics or via graph)?
Hit Error	Records how off the players are with their input. It can show if the game needs offset and their performance.	Recorded the pressed input time and minus with the note time.	Result.HitError	A histogram showing the spread of the hit error.
Judgement of Note	Shows the count of each judgement	From hit error, record the judgement based on the range of the hit error. Also record the type of the note.	Result.JudgementRecord	A stacked bar chart showing the judgement of each note type. Each bar shows how many judgments there are from miss to

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				c.perfect
Combo over time	Shows how many times players have hit the non miss judgement in a row.	Record the combo and the time of recording in a dict.	Result.Combo	A line graph that shows the combo over time.
Left & Right Input	Shows which side players are likely to press the input from.	Record the side that was pressed from the input processing method.	Result.KeyPressSide	A pie chart splitting between left and right.
Player X position	This can be used to see where player characters usually stay in a line	Record the player X position in an interval	Result.PlayerPosition	A line graph showing player x position over time.

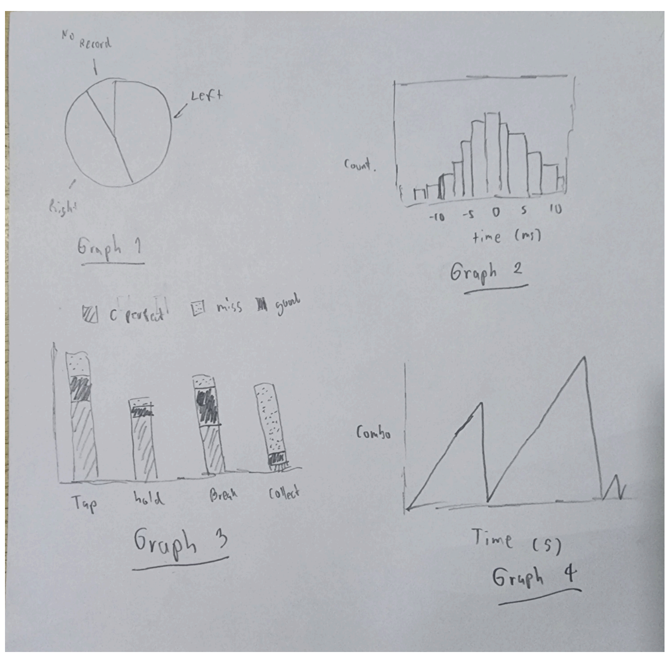
4.2 Data Recording Method

This game will store the player data and play data in a Result class. The game will then write all data from the Result class into a JSON file after the game ends for later review/processing.

4.3 Data Analysis Report

	Feature Name	Graph Objective	Graph Type	X-axis	Y-axis
Graph 1	Side Pressed	To show the proportion between the left side key press and right side key press	Pie Graph	Left side, Right side, No record	-
Graph 2	Hit Error	To show the distribution on how fast/late the player pressed the input to the note.	Histrogram	Hit error milliseconds	Count of the hit error in that range

	Feature Name	Graph Objective	Graph Type	X-axis	Y-axis
Graph 3	Judgement of Note	Shows how many notes of each type there are and the judgement level proportion.	Stacked Bar Chart	Types of note	Count of each judgement level.
Graph 4	Combo over time	Shows how many combos were made over time and when did the combo break.	Line Graph	Time since game start	Combo number



(Hand drawn example of the graph)

A table will show the following data:

Song-Chart Name	Player Name	Total Score Percentage	Max Combo

The table can be used to compare the score between the player and song-chart.

5. Project Planning Timeline

Week	Week	Task
1	8	Proposal submission / Project initiation
2	9	Editor/Main program start
3	10	Editor program done
4	11	Charting/Main program/Playtesting
5	12	Charting/Main program done/Playtesting
6	13	Charting/Bug Fixing/Statistic Analysis
7	14	Submission week (Draft)

6. Document version

Version: 2.1
Date: 5 April 2026
Editor: Pichaya Yawiset 6810545824

7. Glossary

- Danmaku : A game genre that is characterized by the overwhelming obstacle for players to avoid.

- Chart : The design pattern of the action for the song. The action can be anything, such as note, obstacle, etc.